FACULDADE DE DESIGN,
TECNOLOGIA E COMUNICAÇÃO
DA UNIVERSIDADE EUROPEIA

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MASTER'S IN CREATIVE COMPUTING
AND ARTIFICIAL INTELLIGENCE

EEG-Based Remote Car Control Using the Emotiv EPOC X:

System Development and Experimental Evaluation

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Dissertação apresentada ao IADE - Faculdade de Design, Tecnologia e Comunicação da Universidade Europeia, para cumprimento dos requisitos necessários à obtenção do grau de Mestre em Creative Computing and Artificial Intelligence realizada sob a orientação científica da Doutora Pamela Teubig, professora assistente do *IADE* e de Doutor José Graça, professor do IADE.

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Keywords *[Keywords to add.]*

abstract *[Text to write.]*

Glossary

Term *[Definition to write.]*

List of Abbreviations

ABBR *[Meaning to write.]*

Contents

Glossary **1**

List of Abbreviations **2**

List of Figures **6**

List of Tables **7**

1 Introduction **8**
1.1 Background and Context 8
1.2 Motivation 8
1.3 Problem Statement 8
1.4 Research Gap 8
1.5 Research Question 8
1.6 Main Research Objective 8
1.7 Specific Objectives 9
1.8 Research Hypotheses 9
1.9 Scope and Limitations 9
1.10 Expected Contributions 9
1.11 Dissertation Structure 9

2 Brain-Computer Interfaces and EEG-Based Control **10**
2.1 Introduction 10
2.2 Fundamentals of Brain-Computer Interfaces 11
2.3 EEG Technology and Brain Signal Acquisition 11
2.4 EEG-Based BCI Paradigms 11
2.5 Emotiv EPOC X: Architecture and Capabilities 11
2.6 Emotiv Mental Commands: Principles and Limitations 11
2.7 Consumer-Grade EEG Devices in BCI Applications 11
2.8 Machine Learning Approaches for EEG Classification 11
2.9 EEG-Based Control of Vehicles and Mobile Robots 12
2.10 Challenges of Real-Time BCI Control 12
2.10.1 Command Detection and Classification Reliability 12
2.10.2 Response Latency 12
2.10.3 False Activations and Missed Commands 12
2.10.4 Cognitive Workload and User Adaptation 12
2.11 Safety and Reliability in BCI-Driven Systems 12
2.12 Existing Frameworks and Related Work 12
2.13 Research Gap Summary 13

3 Development of an Emotiv EPOC X-Based Brain-Controlled Car System **14**
3.1 Introduction 14
3.2 System Requirements 14
3.3 Overall Architecture 14

3.4 Emotiv EPOC X and Brain Activity Acquisition 14
3.5 Car Construction and Integration of Electronic Components . . . 14
3.6 Microcontroller, Firmware and Motor Control 14
3.7 Development of the Python Framework 15
3.8 Communication Between the Python Framework and the Microcontroller 15
3.9 Data Processing and Identification of Commands 15
3.10 Mapping Commands to Car Movements 15
3.11 Real-time Control and Safety Mechanisms 15
3.12 Data Logging and System Monitoring 15
3.13 Technical Testing and Integration Validation 16
3.14 Chapter Summary 16

4 Research Methodology and Experimental Evaluation 17
4.1 Research Design 17
4.2 Experimental Setup and Equipment 17
4.3 Car Construction and Electronics Integration 17
4.4 Participants and Ethical Considerations 17
4.5 Experimental Environment 17
4.6 Command Training and Calibration 17
4.7 Experimental Tasks and Protocol 18
4.8 Pilot Study 18
4.9 Main Experimental Sessions 18
4.10 Data Collected and Availability 18
4.11 Evaluation Metrics 18
4.11.1 Command Success Rate 18
4.11.2 False Activation Rate 18
4.11.3 Missed Command Rate 18
4.11.4 Command Detection Latency 19
4.11.5 End-to-End Response Time 19
4.11.6 Task Completion Rate 19
4.11.7 Performance Variation Across Sessions 19
4.12 Data Analysis Methods 19
4.13 Optional Investigation of a Custom Machine Learning Approach 19
4.14 Validity, Reliability, and Reproducibility 19
4.15 Chapter Summary 20

5 Results 21
5.1 Introduction 21
5.2 Overview of Collected Experimental Data 21
5.3 Performance of BCI Command Execution 21
5.4 Success Rates and Missed Commands 21
5.5 False Activation Results 21
5.6 Detection Latency and End-to-End Response Time 21
5.7 Results of Car Control Tasks 21
5.8 Performance Variation across Sessions 22

5.9 Results of a Custom Machine Learning Model 22
5.10 Summary of Findings 22

6 Discussion **23**
6.1 Introduction 23
6.2 Interpretation of Experimental Results 23
6.3 Evaluation against Research Objectives 23
6.4 Reliability of BCI-Based Car Control 23
6.5 Latency and Suitability for Real-Time Control 23
6.6 Trade-offs between Speed and Reliability 23
6.7 Implications for Consumer-Grade EEG Applications 23
6.8 Comparison with Previous Studies 24
6.9 Technical and Methodological Limitations 24
6.10 Potential Improvements and Future Work 24
6.11 Chapter Summary 24

7 Conclusion and Future Work **25**
7.1 Introduction 25
7.2 Summary of the Research 25
7.3 Main Contributions 25
7.4 Extent to Which Objectives Were Achieved 25
7.5 Limitations of the Study 25
7.6 Recommendations for Future Research 25
7.7 Final Remarks 25

References **26**

Appendices **27**
Appendix A — Experimental Protocol
Appendix B — Participant Information and Informed Consent Documents
Appendix C — Supplementary Experimental Results
Appendix D — System Architecture and Technical Configuration
Appendix E — Source Code and Repository Information
Appendix F — Instructions for Replicating the Experimental Data

List of Figures

List of Tables

1 Introduction

[Aim: open the dissertation in one or two short paragraphs. Cover: the topic of EEG-based control of a physical device; the Emotiv EPOC X and the remote car as the study platform; the open-source Python framework as the main output.]

1.1 Background and Context

[Aim: give the reader what they need to follow the problem. Cover: what brain-computer interfaces are; non-invasive EEG and the arrival of consumer headsets; vehicle and robot control as an application. Keep this brief and leave the detail to Chapter 2.]

1.2 Motivation

[Aim: explain why the work is worth doing now. Cover: hands-free and assistive control as a real need; lower cost of consumer EEG hardware; the lack of open, reproducible tooling; your own academic interest in the topic.]

1.3 Problem Statement

[Aim: define the problem precisely. Cover: mental commands from consumer EEG are noisy and vary between users; controlling a physical car needs commands that are dependable, timely and safe; it is unclear how far the EPOC X meets those demands. End with a single sentence stating the problem.]

1.4 Research Gap

[Aim: name the gap in brief. Cover: what existing work has shown; what remains missing, such as open frameworks or evaluations reporting reliability and latency together on a physical vehicle. Confirm each claim against the literature; the full evidence belongs in Section 2.13.]

1.5 Research Question

[Aim: state one main question that the experiment can answer. Starting point: to what extent can mental commands from the Emotiv EPOC X, processed by an open-source Python framework, reliably control a remote car in real time? Add sub-questions only if each maps to a metric in Section 4.11.]

1.6 Main Research Objective

[Aim: state the overall objective in one sentence. Starting point: to design, build and experimentally evaluate a brain-controlled car system, driven by Emotiv EPOC X mental commands through an open-source Python framework.]

1.7 Specific Objectives

[Aim: break the main objective into a numbered list. Cover: review the literature; build the car and its electronics; develop the Python framework and integrate it with the car; add safety mechanisms; define the protocol and metrics; run the experiments; analyse the results; publish the code. Start each with a verb, keep each measurable and note the chapter that delivers it, because Section 7.4 reports back on this list.]

1.8 Research Hypotheses

[Include only if applicable. Aim: state testable hypotheses tied to the metrics in Section 4.11. Possible forms: H1, command success rate reaches a stated threshold; H2, end-to-end response time stays within a stated limit; H3, performance changes across sessions. Justify each threshold from the literature or the pilot study, and give null forms if you use inferential statistics.]

1.9 Scope and Limitations

[Aim: set the boundaries of the study. Cover: what is included, such as one headset, Emotiv's built-in detection, one purpose-built car and a controlled environment; what is excluded, such as clinical users and real road use; known constraints such as sample size and time.]

1.10 Expected Contributions

[Aim: say what the work adds. Cover: the brain-controlled car system; the open-source framework; a reusable experimental protocol; a defined set of real-time performance metrics; evidence on the feasibility of consumer EEG control. Separate practical from academic contributions, and revisit them in Section 7.3.]

1.11 Dissertation Structure

[Aim: map the document. Cover: one sentence for each of Chapters 2 to 7 and the Appendices. Write this section last.]

2 Brain-Computer Interfaces and EEG-Based Control

[Aim: open the literature review in one short paragraph. Cover: the subject of the chapter and how it supports the research question.]

2.1 Introduction

A Brain-Computer Interface (BCI), also referred as a Brain-Machine Interface (BMI), is a system that enables direct communication between the human brain and digital devices. Its main purpose is to allow individuals to control computers or other devices using their brain activity. By interpreting neural signals generated by the brain, BCI technology provides an alternative way for people, both with and without physical disabilities, to interact with digital devices.

Brain-Computer Interface systems can be divided into three main categories based on how brain activity is recorded: invasive, partially invasive and non-invasive systems. Invasive BCI systems require surgery to place electrodes directly inside brain tissue, allowing neural signals to be recorded close to their source. Although this approach can provide high-quality neural recordings, it involves surgical risks. Partially invasive systems, such as electrocorticography (ECoG), use electrodes placed on the surface of the cerebral cortex without penetrating brain tissue. In contrast, non-invasive BCI systems record brain activity without surgery, using sensors positioned outside the skull. These approaches are generally safer and easier to implement, although their recordings can present limitations related to signal quality and susceptibility to noise.

Different techniques can be used to acquire brain activity, depending on the type of BCI system. Electroencephalography (EEG) is one of the most commonly used non-invasive techniques and records electrical potentials generated by neural activity through electrodes positioned on the scalp. In contrast, electrocorticography (ECoG) requires surgical intervention to position electrodes directly on the surface of the cerebral cortex. Although both techniques record electrical activity associated with brain function, their electrode placement results in differences in signal quality and spatial resolution. Another non-invasive technique is magnetoencephalography (MEG), which measures the magnetic fields generated by neural activity rather than electrical potentials.

A BCI system operates through three main stages. The first stage is signal acquisition, in which electrical signals generated by brain activity are captured using the selected recording technology. The second stage is signal processing, where the acquired signals are preprocessed to reduce unwanted noise and relevant features are extracted to prepare them for classification. The third stage is data manipulation, in which the processed information is translated into a meaningful format that can be interpreted and used by the final output device, such as a computer or another external device.

2.2 Fundamentals of Brain-Computer Interfaces

[Aim: define the field. Cover: what a BCI is; the standard pipeline of acquisition, processing, translation and output; invasive versus non-invasive; active, reactive and passive BCIs; main application areas. Cite foundational sources.]

2.3 EEG Technology and Brain Signal Acquisition

[Aim: explain how EEG works and where it falls short. Cover: what EEG measures; frequency bands; electrode placement; wet versus dry or saline sensors; artefacts from muscle and eye movement; low spatial resolution and signal-to-noise problems.]

2.4 EEG-Based BCI Paradigms

[Aim: compare the main ways to generate commands. Cover: motor imagery; P300; SSVEP; other mental tasks; facial signals as non-brain alternatives. Compare training needs, command rate and suitability for continuous control, ideally in a table, and say which paradigm this work uses and why.]

2.5 Emotiv EPOC X: Architecture and Capabilities

[Aim: describe the device used in the study. Cover: channels and sensor layout, sampling rate, resolution, connectivity, sensors and software. Take every specification from Emotiv's documentation and cite it; also report independent validation studies against research-grade EEG.]

2.6 Emotiv Mental Commands: Principles and Limitations

[Aim: explain the detection feature your framework depends on. Cover: how users train commands; the neutral state; how many commands can be active; the output of command and power values; what is and is not public about the algorithm. Note the consequences of treating it as a black box.]

2.7 Consumer-Grade EEG Devices in BCI Applications

[Aim: place the EPOC X among similar devices. Cover: other consumer headsets; cost and usability benefits; accuracy limits compared with research systems; reported use in BCI studies.]

2.8 Machine Learning Approaches for EEG Classification

[Aim: review how EEG is classified. Cover: feature extraction; classical models such as LDA and SVM; deep learning such as CNN-based models; subject-dependent versus subject-independent training; small dataset and overfitting problems. Link this to the optional work in Section 4.13.]

2.9 EEG-Based Control of Vehicles and Mobile Robots

[Aim: review the closest applied work. Cover: studies of EEG-controlled cars, wheelchairs and robots; for each, the device, paradigm, number of commands, participants and results. A comparison table works well here and supports Section 6.8.]

2.10 Challenges of Real-Time BCI Control

[Aim: introduce the challenges that shape the framework and metrics. Cover: why online control is harder than offline classification, and how each subsection below links to a metric in Section 4.11.]

2.10.1 Command Detection and Classification Reliability

[Aim: review how reliable command detection is in practice. Cover: reported accuracy and success rates; variation between users and sessions; differences between offline and online figures; how studies measure reliability.]

2.10.2 Response Latency

[Aim: explain the delays in a BCI loop. Cover: the sources of delay, including signal windowing, processing, transmission and actuation; reported latencies; what counts as acceptable for vehicle control.]

2.10.3 False Activations and Missed Commands

[Aim: explain the two main error types. Cover: the difference between false activations and missed commands; why false activations matter most for a moving car; thresholding and the trade-off between the two errors.]

2.10.4 Cognitive Workload and User Adaptation

[Aim: cover the human side of control. Cover: fatigue and mental effort; BCI inefficiency, where some users cannot gain control; learning across sessions; how workload is measured, such as NASA-TLX.]

2.11 Safety and Reliability in BCI-Driven Systems

[Aim: review safety in brain-controlled devices. Cover: why noisy input calls for protection; emergency stops and fail-safe states; shared control; watchdogs and timeouts; ethical duties to users. Use this to justify the design in Section 3.11.]

2.12 Existing Frameworks and Related Work

[Aim: review available software. Cover: open-source BCI tools and Emotiv-based projects; what they support; their licence, maintenance and limitations. State what an existing tool would not give this project, justifying a new framework.]

2.13 Research Gap Summary

[Aim: close the chapter with the evidence for the gap. Cover: the main findings of the review; the specific gap this study addresses, more fully than Section 1.4; how the findings shape the framework design and experimental method.]

3 Development of an Emotiv EPOC X-Based Brain-Controlled Car System

[Aim: open the chapter in one short paragraph. Cover: the purpose of the brain-controlled car system and how this chapter answers the design objectives from Section 1.7.]

3.1 Introduction

[Aim: outline the chapter. Cover: the design goals, namely real-time control, safety, data capture and openness; the order of the sections that follow.]

3.2 System Requirements

[Aim: list what the system must do. Cover: functional requirements, such as acquiring commands, mapping them, driving the car, stopping and logging; non-functional requirements, such as latency, reliability and portability. Give each an identifier, such as FR1 or NFR1, so Section 3.13 can test it.]

3.3 Overall Architecture

[Aim: show the whole system at a glance. Cover: an architecture diagram from headset through the Python framework and microcontroller to the motors; the main modules and their responsibilities; data flow; why the system is modular. Place the diagram in figures/ch3.]

3.4 Emotiv EPOC X and Brain Activity Acquisition

[Aim: explain how brain activity enters the system. Cover: headset fitting and contact quality; authentication and app credentials; session creation; subscribing to the data stream; handling errors and disconnections. Follow the Cortex API documentation for the version you used, and never show real credentials.]

3.5 Car Construction and Integration of Electronic Components

[Aim: describe the design of the car. Cover: chassis, motors and steering; motor driver, power supply and wireless module; wiring, with a diagram and photos; a component table with part names and quantities; why each part was chosen. Section 4.3 records the car as used in the experiments, so keep the design reasoning here.]

3.6 Microcontroller, Firmware and Motor Control

[Aim: explain the embedded software. Cover: the microcontroller chosen and why; firmware structure and development tools; how commands become motor

signals, such as PWM speed control; start-up state and stopping behaviour. Use a short code listing if useful.]

3.7 Development of the Python Framework

[Aim: explain how the framework was built. Cover: Python version, libraries and module structure; concurrency, such as asyncio or threads; configuration; a table of software versions. Also cover its open-source design: repository structure, licence, dependency management, installation and how to extend it to other devices, with full details in Appendix E.]

3.8 Communication Between the Python Framework and the Microcontroller

[Aim: describe the link to the car. Cover: the wireless technology used; the message format; acknowledgements and retries; behaviour when the link drops. Explain why you chose this option over the alternatives.]

3.9 Data Processing and Identification of Commands

[Focus on Emotiv Mental Commands if that is the option you choose. Aim: explain what happens to a command on arrival. Cover: the received action and power values; timestamps; threshold filtering; smoothing or debouncing, such as requiring a command to persist. Justify each parameter and show a short code listing if useful.]

3.10 Mapping Commands to Car Movements

[Aim: explain how brain commands become car actions. Cover: the mapping table, such as push to forward; the role of neutral; handling of conflicting commands; state logic, ideally as a state diagram.]

3.11 Real-time Control and Safety Mechanisms

[Aim: explain how the system stays timely and safe. Cover: the control loop and update rate; buffering and avoiding stale commands; where timestamps are captured for latency measurement; a manual emergency stop independent of the EEG channel; stopping on lost signal or a timeout watchdog; default to stopped; speed limits. Show which risks each mechanism covers, drawing on Section 2.11.]

3.12 Data Logging and System Monitoring

[Aim: explain what the system records and displays. Cover: logged fields, such as timestamps, commands, power values and car actions; file format; session identifiers; how participant data stays pseudonymised; live monitoring of signal quality, connection and car state during sessions. Make sure the logs can produce every metric in Section 4.11.]

3.13 Technical Testing and Integration Validation

[Aim: show the system works before people use it. Cover: unit tests; bench tests of motors and firmware; integration tests from framework to car; a simulated command source for tests without a headset; timing and emergency stop tests; a table linking requirements from Section 3.2 to test results.]

3.14 Chapter Summary

[Aim: close the chapter. Cover: what was built, how it meets the requirements, and how it serves as the instrument for the evaluation in Chapter 4.]

4 Research Methodology and Experimental Evaluation

[Aim: open the chapter in one short paragraph. Cover: how the evaluation will answer the research question and test any hypotheses; the order of the sections that follow.]

4.1 Research Design

[Aim: name and justify the design. Cover: quantitative experimental evaluation, or a mixed approach if you add questionnaires; within-subject or repeated measures as applicable; independent and dependent variables; why this design fits the question.]

4.2 Experimental Setup and Equipment

[Aim: let another researcher rebuild the setup. Cover: headset and sensor preparation; the car; the computer and software versions; the physical arrangement, with a photo or diagram; the procedure for checking signal contact quality.]

4.3 Car Construction and Electronics Integration

[Aim: record the car as used in the experiments. Cover: the final configuration, with a photo; any changes made after Chapter 3; how consistent behaviour was checked before each session, such as battery level and motor response. Refer to Section 3.5 for the design rather than repeating it.]

4.4 Participants and Ethical Considerations

[Aim: describe who took part and how they were protected. Cover: recruitment; inclusion and exclusion criteria; number of participants and why; informed consent, with forms in Appendix B; ethics approval from the institution; data protection under GDPR; safety and the right to withdraw. If approval is pending, say so and do not state it as granted.]

4.5 Experimental Environment

[Aim: describe where the tests took place. Cover: the room or track; layout and dimensions; lighting, noise and sources of electrical interference; how conditions were kept consistent between sessions.]

4.6 Command Training and Calibration

[Aim: explain how participants prepared. Cover: which commands were trained; the training tool; number and length of training rounds; the criterion for being ready; how the same procedure was kept for everyone.]

4.7 Experimental Tasks and Protocol

[Aim: give the step-by-step procedure. Cover: each task, such as cued single commands and driving a set course; trial numbers and order; cue timing; rest breaks; rules for restarting or aborting. Put the full protocol in Appendix A.]

4.8 Pilot Study

[Aim: explain how the method was tested first. Cover: who took part; what you tried; what you changed as a result, such as thresholds, task length or instructions. Keep pilot data separate from the main results.]

4.9 Main Experimental Sessions

[Aim: describe how the main study ran. Cover: number of sessions and spacing between them; session structure and timing; who was present; how sessions were recorded; any deviations from the protocol.]

4.10 Data Collected and Availability

[Aim: state what data exists and who can access it. Cover: raw EEG if accessible, recognised commands, timestamps and the car's responses to tasks; file organisation and naming; pseudonymisation; secure storage and retention; what can be shared openly, with reproduction details in Appendix F. Check whether your Emotiv licence gives access to raw EEG.]

4.11 Evaluation Metrics

[Aim: define how performance is measured. Cover: how the metrics link to the hypotheses and objectives; shared definitions of trial, command window and correct response. For each metric below give a formula, the unit and the logged data used.]

4.11.1 Command Success Rate

[Aim: define this metric. Cover: correct commands executed as a share of commands cued; what counts as correct; the time window allowed.]

4.11.2 False Activation Rate

[Aim: define this metric. Cover: commands detected when none was intended, or a different one was; whether it is expressed per minute or as a share of trials, and why.]

4.11.3 Missed Command Rate

[Aim: define this metric. Cover: cued commands not detected within the time window, as a share of commands cued; how it relates to the success rate.]

4.11.4 Command Detection Latency

[Aim: define this metric. Cover: time from cue to detection by the framework; where each timestamp comes from; clock synchronisation; timing accuracy and how you verified it.]

4.11.5 End-to-End Response Time

[Aim: define this metric. Cover: time from cue to physical car response; how car movement is timestamped, such as onboard feedback or video; the breakdown into detection, processing and transmission delay.]

4.11.6 Task Completion Rate

[Aim: define this metric. Cover: tasks finished as a share of attempted tasks; the criteria for success and failure, such as time limits or leaving the course; any completion time measures.]

4.11.7 Performance Variation Across Sessions

[Aim: define how change is measured. Cover: the metrics compared across sessions; how learning, fatigue and variation between participants are examined; how you handle missing sessions.]

4.12 Data Analysis Methods

[Aim: explain the analysis before it is run. Cover: data cleaning and exclusion rules; descriptive statistics; statistical tests linked to each hypothesis, with assumption checks; significance level; effect sizes; software used. Choose tests suited to your sample size.]

4.13 Optional Investigation of a Custom Machine Learning Approach

[Aim: describe the optional comparison, if it is carried out. Cover: why a custom classifier could improve on built-in detection; available data; preprocessing and features; candidate models; validation that avoids leakage between sessions; comparison with Emotiv's detection. If it is not done, say so and remove Section 5.9.]

4.14 Validity, Reliability, and Reproducibility

[Aim: assess the method's strengths and threats. Cover: internal validity, such as learning and fatigue effects; external validity, such as sample and setting; reliability of measurement; how the open code, protocol and data allow reproduction.]

4.15 Chapter Summary

[Aim: close the chapter. Cover: the design, metrics and analysis in brief, leading into the results.]

5 Results

[Aim: open the chapter in one short paragraph. Cover: that it reports the findings without interpretation, which belongs in Chapter 6. Use only measured data.]

5.1 Introduction

[Aim: outline the chapter. Cover: how results are organised and how they follow the metrics in Section 4.11.]

5.2 Overview of Collected Experimental Data

[Aim: describe the data obtained. Cover: participants and sessions completed; total trials; exclusions and reasons; data quality issues; deviations from the protocol. A summary table works well.]

5.3 Performance of BCI Command Execution

[Aim: report how well commands were detected. Cover: overall and per-command results; a confusion matrix of cued versus detected commands; variation between participants.]

5.4 Success Rates and Missed Commands

[Aim: report these two rates. Cover: mean and spread for each command and participant; a table and a chart; the result of any hypothesis test on H1.]

5.5 False Activation Results

[Aim: report unintended commands. Cover: rates by participant and condition; which commands were most often triggered wrongly; any safety events or emergency stops.]

5.6 Detection Latency and End-to-End Response Time

[Aim: report the timing results. Cover: mean, median, spread and outliers; distribution plots; the breakdown of delay by stage; the result of any hypothesis test on H2.]

5.7 Results of Car Control Tasks

[Aim: report task performance. Cover: completion rates by task and participant; completion times; reasons for failure.]

5.8 Performance Variation across Sessions

[Aim: report changes over time. Cover: the metrics by session, shown in a line chart or table; participant differences; the result of any hypothesis test on H3.]

5.9 Results of a Custom Machine Learning Model

[Aim: report the optional comparison, if carried out. Cover: model performance with validation method; comparison with built-in detection using the same data; any offline-only caveat. Remove this section if the work was not done.]

5.10 Summary of Findings

[Aim: close the chapter with the main results. Cover: the key figures for each metric; the outcome for each hypothesis, as supported or not supported, without explaining why.]

6 Discussion

[Aim: open the chapter in one short paragraph. Cover: how it interprets the results against the research question.]

6.1 Introduction

[Aim: outline the chapter. Cover: restate the research question and the order of the sections that follow.]

6.2 Interpretation of Experimental Results

[Aim: say what the results mean. Cover: the main findings and their likely explanations; unexpected results; whether each hypothesis held and why. Distinguish what the data show from what you infer.]

6.3 Evaluation against Research Objectives

[Aim: assess the work against its objectives. Cover: for each objective in Section 1.7, the supporting evidence; a table may help. Answer the research question directly.]

6.4 Reliability of BCI-Based Car Control

[Aim: judge whether commands are dependable enough. Cover: what success, missed command and false activation rates mean for real use; which factors affected reliability, such as user, command or training.]

6.5 Latency and Suitability for Real-Time Control

[Aim: judge whether the system is fast enough. Cover: what the delays mean for driving a car; which stage contributes most delay; which parts could improve and which depend on Emotiv's detection.]

6.6 Trade-offs between Speed and Reliability

[Aim: discuss the central design tension. Cover: how stricter thresholds and smoothing reduce errors but add delay; what your data shows about this; the balance you recommend, and for what use.]

6.7 Implications for Consumer-Grade EEG Applications

[Aim: look beyond this car. Cover: what the results suggest for assistive technology, robotics and other uses of consumer EEG; where such devices are and are not appropriate, including safety-critical limits.]

6.8 Comparison with Previous Studies

[Aim: place results in the literature. Cover: comparison with the studies in Section 2.9, by device, paradigm and metric; agreements and differences; why direct comparison may be limited.]

6.9 Technical and Methodological Limitations

[Aim: assess the study honestly. Cover: sample size and participant mix; controlled environment; black-box detection; hardware and timing measurement limits; how each may have affected the results.]

6.10 Potential Improvements and Future Work

[Aim: suggest practical improvements to this system. Cover: changes to the framework, control logic, safety and protocol, each linked to a finding or limitation. Broader research directions belong in Section 7.6.]

6.11 Chapter Summary

[Aim: close the chapter. Cover: the main interpretation and its significance, leading into the conclusion.]

7 Conclusion and Future Work

[Aim: open the chapter in one short paragraph. Cover: that it brings the work together. Introduce no new results or sources.]

7.1 Introduction

[Aim: outline the chapter. Cover: restate the research question and the order of the sections that follow.]

7.2 Summary of the Research

[Aim: retell the work briefly. Cover: the problem; what was built; how it was tested; the main findings.]

7.3 Main Contributions

[Aim: state what the work delivered. Cover: the actual outcomes against the expected contributions in Section 1.10, including the framework, protocol, evidence and any data released.]

7.4 Extent to Which Objectives Were Achieved

[Aim: close the loop on the objectives. Cover: each objective from Section 1.7, marked as met, partly met or not met, with a short justification; a direct answer to the research question.]

7.5 Limitations of the Study

[Aim: state the limits on the conclusions. Cover: a brief summary of Section 6.9 and how far the findings can be generalised.]

7.6 Recommendations for Future Research

[Aim: suggest what other researchers should do next. Cover: larger and more diverse samples; custom classifiers; other paradigms or devices; less controlled environments; long-term studies. Base each on your findings.]

7.7 Final Remarks

[Aim: end the dissertation in one short paragraph. Cover: the wider significance of the work and the main message for the reader.]

References

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Appendices

Appendix A — Experimental Protocol

[Aim: give the complete protocol. Include: step-by-step session script; participant instructions; task definitions and cue timing; session checklist; abort and emergency procedures.]

Appendix B — Participant Information and Informed Consent Documents

[Include only if applicable. Aim: provide the ethics documents. Include: participant information sheet; blank consent form; questionnaires; the ethics approval reference, if granted. Exclude anything that identifies a participant.]

Appendix C — Supplementary Experimental Results

[Aim: hold detail too long for Chapter 5. Include: per-participant and per-session tables; extra charts; full statistical outputs.]

Appendix D — System Architecture and Technical Configuration

[Aim: give the full technical record. Include: detailed diagrams; car wiring or hardware details; configuration parameters and thresholds; message format; software versions.]

Appendix E — Source Code and Repository Information

[Aim: point readers to the code. Include: repository URL; the release or commit used in the study; licence; folder structure; any key listings. Never include API credentials.]

Appendix F — Instructions for Replicating the Experimental Data

[Aim: let others repeat the analysis. Include: where the anonymised data is stored and how to access it; data dictionary; installation steps; commands that reproduce each result.]

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